Raghavan 2021 Pancreatic Ductal Adenocarcinoma — analysis summary

PDAC (metastatic) · Seq-Well · 7,830 malignant cells (21 samples, gap-threshold ≥120)

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

21/23 samples (≥120-cell gap threshold) intermix on one continuous manifold — iLISI median 1.00→7.04 of 21 (θ sweep 2/5/10, monotone), 0 detached islands, UMAP aspect 1.10 (one blob); both compartments (Classical-ductal 3,092 / Cycling 4,738) span all 21 samples.

Gate 2 — module geography → GO (non-branching)

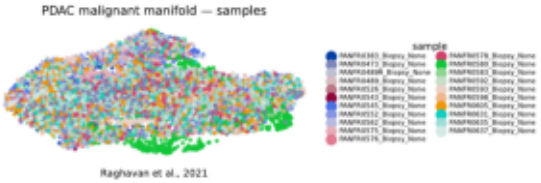
Single pSom→Crest axis: somitogenesis high at the origin declines along pseudotime (ρ=−0.64) while neural crest rises to a terminal readout (ρ=+0.48). Gast/pSom co-express (joint-high enrichment 1.13 >1 — no competing arms); Crest anti-correlated with both (r=−0.75 Gast, −0.65 pSom); joint pSom&Crest-high depleted (0.23) — an inverse gradient, not a hub.

Gate 3 — hub cross-repression → NA

Non-branching manifold (principal curve, 2 tips / 0 forks; PPT forks are blob over-fit, not dev-signature arms). No bifurcation and no Gast/pSom co-expression hub — no branch pair to test for mutual exclusivity; the correlation / modules notebooks are not produced.

STEP FIGURES

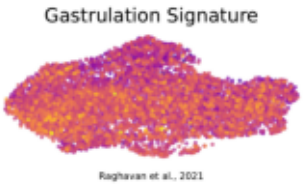
Gate 1: integration



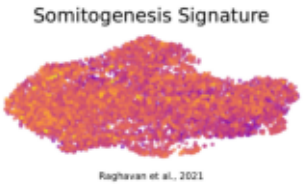
Trajectory / root



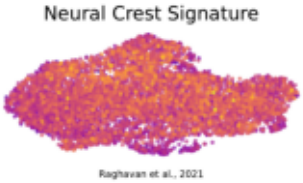
Gast



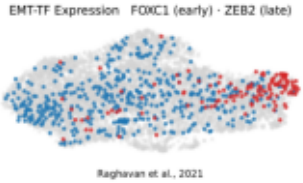
pSom



Crest



EMT-TFs



Dataset. PDAC (Raghavan et al., 2021), Seq-Well — metastatic biopsies, 24 samples / 24,412 malignant cells (master scope). After the ≥120-cell gap threshold, 21 samples / 7,830 cells enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict. Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous somitogenesis→neural-crest axis rooted at the Gast/pSom-high origin. Gast and pSom co-express (no bifurcation), so Gate 3 has no hub to test (NA). Crest is the terminal readout, not Gast-enriched (no inversion flag). Raw module means (Gast −0.011 / pSom +0.072 / Crest −0.024) match the failed prior run, but the call turns on geometry: a coherent single pSom→Crest gradient with a terminal crest readout — GO, not the prior undecided call. Proliferation is orthogonal to the axis (module–cycling |ρ|≤0.19). EMT-TFs FOXC1 (early/root, 23.6%) → ZEB2 (late/terminus, 6.4%); canonical SNAI1/TWIST1 near the detection floor. G1 GO / G2 GO / G3 NA.

Parameter choices

Min cells/sample	≥120 (gap 120→45, r=2.67) → 21/23	**Integration**	Harmony θ=10; n_pcs=20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Geometry**	Linear (non-branching); 2 tips / 0 forks
Annotation	Moffitt classical/basal + Raghavan mp	**EMT-TFs**	FOXC1 (early) · ZEB2 (late)